Plasma Surface Modification of PTFE: A Detailed Review of Mechanisms, Process Parameters, and Performance Enhancement

Abstract

Polytetrafluoroethylene (PTFE) possesses exceptional chemical inertness, thermal stability, and low surface energy, yet its poor adhesion and wettability severely limit its application in composites and bonded assemblies. Plasma surface modification has emerged as a versatile, environmentally friendly technique to overcome these limitations without altering bulk properties. This paper systematically reviews the mechanisms, process parameters, and performance outcomes of plasma treatment on PTFE. The primary mechanisms involve defluorination through C–F bond scission by energetic species and VUV radiation, introduction of oxygen- and nitrogen-containing polar functional groups, and surface etching that increases roughness. Atmospheric pressure plasma systems, including multi-hollow surface dielectric barrier discharge (MSDBD) and gliding arc jets, offer advantages over low-pressure systems in terms of scalability and continuous processing. Oxygen plasma treatment increases PTFE surface polarity to 0.44, significantly higher than polyethylene (0.26). Argon and nitrogen plasma treatments improve adhesive strength by 17.9% and 20.0%, respectively, and reduce friction coefficient from 0.34 to 0.16–0.19. Plasma-functionalized PTFE micropowders in PESU composites exhibit 75% higher force absorption (3.3 kN vs. 0.96 kN) and nearly half the wear rate. A critical challenge is the aging effect, with hydrophobic recovery occurring predominantly within 7–14 days post-treatment. Strategies to mitigate aging include plasma-induced graft polymerization and low-temperature storage.

Keywords: PTFE; Plasma treatment; Surface modification; Defluorination; Adhesion; Wettability

1. Introduction

Polytetrafluoroethylene (PTFE) is renowned for its unique combination of properties: chemical inertness, high thermal stability, low surface tension, and excellent mechanical stability . These characteristics make PTFE indispensable in applications ranging from chemical processing equipment to self-lubricating bearings and high-frequency electronics. However, the same

perfluorinated molecular structure that confers these advantages also creates a fundamental limitation: PTFE exhibits extremely poor adhesion and wettability, making it difficult to bond, coat, or incorporate into composite materials .

Conventional surface modification techniques for PTFE, such as sodium-naphthalene complex treatment, suffer from significant drawbacks. The wet chemical process can create harmful residual substances and cause loss of optical and mechanical properties due to harsh etching . Treated PTFE turns dark and becomes brittle . Radiation grafting increases roughness but involves long-term radiation exposure with associated health concerns .

Plasma surface modification has emerged as the most promising alternative. Plasma treatment offers several advantages: it is environmentally friendly, efficient, and can modify the surface without affecting bulk properties . The technique can be applied under low-pressure conditions or at atmospheric pressure, with the latter offering scalability and continuous processing capabilities .

This paper provides a comprehensive review of plasma surface modification of PTFE, covering the fundamental mechanisms, process parameter optimization, characterization methods, and performance outcomes reported in the literature.

2. Fundamental Mechanisms of Plasma-PTFE Interaction

2.1 C–F Bond Scission and Defluorination

The primary mechanism of plasma modification of PTFE involves the breaking of strong C–F bonds (bond energy ~485 kJ/mol) by energetic plasma species. The flux of plasma species—including electrons, ions, and radicals such as nitrogen and oxygen radicals and hydroxyl—collides with the polymer surface, causing C–F bond scission . Simultaneously, the PTFE surface is exposed to vacuum ultraviolet (VUV) radiation, which also contributes to C–F bond cleavage .

The defluorination process results in a surface depleted of fluorine, with dangling bonds available for reaction with oxygen- and nitrogen-containing species. XPS analysis has quantified this effect: PTFE surfaces treated at lower power present 60.0 ± 4.7 atom% fluorine with an F/C ratio of 1.56, while higher power treatment reduces fluorine to 52.6 ± 0.5 atom% with an F/C ratio of 1.24 . The increase in nitrogen and oxygen content with increasing power suggests that higher power facilitates greater incorporation of these elements onto the PTFE surface .

2.2 Introduction of Polar Functional Groups

Following C–F bond scission, reactive species in the plasma interact with the surface to introduce polar functional groups. ATR-FTIR analysis of plasma-treated PTFE reveals the emergence of new absorption bands: peaks at 2913 cm^{-1} and 2845 cm^{-1} related to C–H stretching indicate alkyl or vinyl group formation from C–F bond cleavage . A peak at 1630 cm^{-1} corresponds to C=C stretching, likely resulting from backbone scission or unsaturation .

After 5 minutes of plasma treatment, more pronounced modifications occur: the original C–F peak range (1200–1140 cm^{-1}) decreases and broadens, possibly due to defluorination or disruption of the fluorocarbon network . The reduced intensity of peaks at 718 cm^{-1} and 1461 cm^{-1}, associated with CF_2 deformation modes, suggests partial replacement of CF_2 groups by oxygen-containing functional groups . In general, plasma treatment introduces oxygen-containing functional groups including carbonyls and hydroxyls on the otherwise chemically inert PTFE surface .

2.3 Surface Etching and Roughening

Plasma treatment also causes physical etching of the PTFE surface through ion bombardment. AFM analysis reveals that SDBD air plasma treatment leads to the formation of microscale roughness on PTFE surfaces . This occurs due to bombardment of the surface by high-energy electrons and ions, causing physical etching and removal of polymer chains .

The degree of surface roughness depends on plasma treatment parameters such as power, treatment time, and gas composition . Interestingly, higher power levels may lead to harsher etching that can smooth out the surface by removing irregularities, while reducing power

decreases the etching rate, allowing surface irregularities to remain or become more pronounced . This roughening increases the surface area and improves mechanical interlocking between PTFE and other materials like coatings, adhesives, or inks .

3. Plasma Treatment Systems and Process Parameters

3.1 Low-Pressure Plasma Systems

Low-pressure plasma systems, operating in the mTorr range (10^{-6} atm), have been extensively used for PTFE modification. These systems include microwave plasma sources such as Plasmodul® and Planartron®, which can generate homogeneous, linearly extended plasmas . The Plasmodul® source allows pre-treatment of PTFE substrates with ammonia plasma, while the Planartron® source generates oxygen and nitrogen plasmas .

A notable advantage of low-pressure systems is the ability to achieve uniform treatment over complex geometries. However, these systems have significant drawbacks: substrate size is restricted to fit within the reactor chamber, specialized vacuum pumping systems are required, and samples must be processed in batches, making the approach less time-effective .

3.2 Atmospheric Pressure Plasma Systems

Atmospheric pressure plasma systems have been developed to overcome the limitations of low-pressure systems. The multi-hollow surface dielectric barrier discharge (MSDBD) system has been successfully applied for modifying PTFE foils . This system offers versatility, robustness, and scalability for industrial applications . Treatment with MSDBD plasma in air, H_2/N_2 gas mixture, and pure oxygen significantly decreases the water contact angle (WCA) of PTFE with a treatment time of only 10 seconds .

The gliding arc jet system represents another atmospheric pressure approach. In this configuration, plasma discharge is displaced by Ar gas flow under an acrylic acid environment,

resulting in PTFE coated with a hydrophilic layer that drastically increases adhesion . The gliding arc is energetic enough to remove and/or cross-link the weak boundary layer and graft polar functionalities onto the PTFE surface .

The helix atmospheric-pressure plasma system provides a uniform plasma discharge ideal for surface treatment of polymeric materials. This system operates at 13.56 MHz RF power and allows coverage of large surface areas .

Table 1: Comparison of Plasma Treatment Systems for PTFE

System Type	Operating Pressure	Advantages	Limitations
Low-pressure microwave	~mTorr	Uniform treatment, well-established	Batch processing, size restrictions
MSDBD atmospheric	Atmospheric	Scalable, short treatment time (10 s)	Less uniform than low-pressure
Gliding arc jet	Atmospheric	Grafting capability, high energy	Localized treatment
Helix atmospheric	Atmospheric	Large area coverage, continuous	Requires optimization

4. Characterization of Plasma-Modified PTFE

4.1 Wettability and Surface Energy

Water contact angle (WCA) measurement is the most widely used method for assessing plasma treatment effectiveness. Untreated PTFE exhibits a highly hydrophobic surface with WCA typically above 110° . Plasma treatment substantially reduces WCA, with all three working gases (air, H_2/N_2 mixture, and O_2) significantly decreasing WCA after only 10 seconds of treatment .

The surface polarity values provide a quantitative measure of functionalization. Untreated PTFE displays virtually zero polarity (≈0.01), indicating a lack of polar functionality . Following plasma treatment, PTFE reaches a significantly higher polarity value of 0.44, compared to 0.26 for polyethylene under the same conditions . This higher value highlights the greater susceptibility of PTFE to plasma-induced functionalization .

4.2 X-Ray Photoelectron Spectroscopy (XPS)

XPS analysis provides quantitative information on surface chemical composition. The incorporation of nitrogen and oxygen increases with plasma power: at lower power, PTFE surfaces present 1.3 ± 0.2 atom% nitrogen and 0.5 ± 0.1 atom% oxygen, while at higher power, these increase to 3.9 ± 0.7 atom% and 1.0 ± 0.5 atom%, respectively . The F/C ratio decreases from 1.56 to 1.24 with increasing power, confirming defluorination .

4.3 Atomic Force Microscopy (AFM)

AFM analysis reveals changes in surface morphology. Lower and higher power treatments increased the average surface roughness (Sa) to 120 ± 50 nm and 110 ± 12 nm, respectively, compared to 79 ± 2 nm for untreated PTFE . These variations in physicochemical properties lead to differences in peel strength .

5. Performance Enhancement through Plasma Treatment

5.1 Adhesive Properties

Plasma treatment dramatically improves the adhesive properties of PTFE. For PTFE/Nomex fabric composites, oxygen plasma treatment enhanced interfacial adhesion between fibers/fabric and resin, resulting in lower friction coefficient and wear volume compared to neat composites .

Argon and nitrogen plasma treatments of hybrid PTFE/Kevlar fabrics improve adhesion between fibers and epoxy by 17.9% (Ar) and 20.0% (N_2) . The treated hybrid fabric composite shows better tribological performance due to strong adhesion of fabric .

A particularly impressive result was achieved with plasma-assisted modification combined with homologous adhesive bonding: peel strength increased nearly 14-fold from 0.0413 to 0.574 N/mm . Molecular dynamics simulations confirmed that the interaction energy at the PTFE/AF interface increased to −114.46 kcal/mol after plasma modification, indicating that increased polar groups significantly improve interfacial interaction .

5.2 Tribological Properties

Plasma treatment significantly improves the tribological performance of PTFE composites. The friction coefficient of treated hybrid PTFE/Kevlar fabric composite is reduced to 0.19 (Ar) and 0.16 (N_2) compared to 0.34 for the original composite . The treated fabric composite shows better wear resistance and self-lubrication performance due to improved wettability and adhesive properties .

For PTFE micropowders in PESU composites, plasma-treated PTFE–PESU composites exhibit nearly 75% higher force absorption capabilities (3.3 kN) compared to pristine PTFE–PESU counterparts (0.96 kN) . Moreover, the plasma-treated composites exhibit a wear rate ($3.42 \pm 0.51 \times 10^{-6}$ mm³/Nm) nearly half that of pristine composites ($5.75 \pm 0.80 \times 10^{-6}$ mm³/Nm) .

The enhanced wear resistance can be attributed to improved interfacial adhesion resulting from the rougher surface and hydrophilic groups introduced onto the PTFE fiber surface by plasma modification . The improved interfacial adhesion between fabric and resin reduces abrasive wear and promotes the formation of a PTFE lubrication film .

5.3 Mechanical Properties

Plasma treatment of fillers can enhance the mechanical properties of PTFE composites. RF plasma treatment of polyoxadiazole (POD) fibers in tetrafluoroethylene medium raises composite density, heat resistance, and mechanical strength . The tensile strength of the 10%-filled composite increases by more than 20%, and elongation at break by approximately 50% compared to composites filled with original fibers . The filler treatment produces a fluoroorganic coating with good wettability by the matrix polymer, which reduces void content and raises adhesion between matrix and fibers .

5.4 Dielectric Properties

An important consideration for electronic applications is the effect of plasma treatment on dielectric properties. Studies on PTFE/AF composites show that plasma mainly acts on the extremely thin surface of the material, with limited influence on bulk properties . The dielectric constant (Dk) of PTFE-P/AF is 2.044 with Df of 0.00041, even better than untreated samples (Dk = 2.055, Df = 0.00044) due to improved interface strength that weakens interface polarization .

6. Aging Effect and Stability of Plasma-Treated PTFE

6.1 Hydrophobic Recovery

A critical limitation of plasma treatment is the aging effect, known as hydrophobic recovery, where the modified surface gradually loses hydrophilicity over time . This phenomenon is attributed to the reorientation of polymeric chains from the treated surface into the bulk, hindering interactions of plasma-introduced functionalities with adhesives . The decrease in adhesion can also be related to desorption of reactive species, moisture absorption, or surface degradation .

Most significant surface recovery occurs in the 7–14 day window after treatment . Contact angle measurements after 30 days show that advancing contact angles are greater than those measured immediately after treatment . For power inputs ≥100 W, they exceed 150°, fulfilling the first

condition for super-hydrophobic surfaces . This recovery is explained by surface adaptation: reptation motion moves some polymer chains from surface to bulk, and chain rotation decreases the density of hydroxyl groups exposed to air .

6.2 Strategies to Mitigate Aging

Several strategies have been developed to mitigate the aging effect. Plasma-induced graft polymerization has been identified as a highly effective method . By generating active sites on the polymer surface, this approach enables covalent attachment of functional monomers, creating durable modifications . Acrylic acid is frequently employed for plasma-induced poly(acrylic acid) graft polymerization on PTFE surfaces .

Low-temperature storage can also mitigate aging effects. Studies have shown that reducing temperature to −10°C at reduced pressure (50 kPa) helps preserve adhesion properties . Low-pressure plasma graft polymerization, in which monomers with unsaturated bonds are polymerized sequentially on the plasma-activated surface, represents another durable modification approach .

7. Conclusions

Plasma surface modification is a highly effective technique for enhancing the surface properties of PTFE while preserving its bulk characteristics. The primary mechanisms involve defluorination through C–F bond scission, introduction of polar functional groups, and surface roughening. Atmospheric pressure plasma systems offer significant advantages over low-pressure systems in terms of scalability and processing efficiency.

Key performance outcomes include:

- Surface polarity increases to 0.44, significantly higher than polyethylene (0.26)

- Adhesive strength improves by 17.9–20.0% with Ar and N_2 plasma

- Friction coefficient reduces from 0.34 to 0.16–0.19

- Force absorption capability increases by 75% in PESU composites

- Wear rate reduces by nearly 50%

- Peel strength can increase up to 14-fold with optimized treatment

The primary limitation is the aging effect, with hydrophobic recovery occurring predominantly within 7–14 days. Mitigation strategies include plasma-induced graft polymerization and low-temperature storage. Future research should focus on developing durable modification approaches that maintain performance over extended service life while retaining the advantageous bulk properties of PTFE.

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